

Toumu’a Lending

Diagram and report brief

Prepared for **Ian Louw** (Visio + written report)

Client: Alice Po’uhila, Toumu’a Money Transfer Ltd • Supervisor: Havea Fonua

Live system: <https://web-production-b5acc.up.railway.app>

Source: <https://github.com/chenmin911130-cyber/toumua-lending>

Your job. Draw the diagrams in Visio and write short captions for the design report. Do not write code. Do not invent extra statuses, interest rates, or actors. Trace the live application and this brief. Send the .vsdx (or PDF export) plus captions to Myo Thant on Friday.

1. What the system is

Toumu’a Lending is a school-built loan workspace for a small Auckland lending office. Staff and customers use a web app. Office staff can also open the same staff screens in a desktop window (Electron). One API and one PostgreSQL database sit behind both clients.

The supported business is:

1. A loan officer and borrower complete an application and record collateral.
2. A valuation officer records a valuation.
3. A manager approves or declines.
4. Collateral is received, inspected, and stored.
5. A cashier disburses the loan, then records repayments and issues receipts.
6. On settlement the collateral is returned. On default it may be sold and the proceeds recorded.

Interest currently uses a **labelled demo policy** (21% p.a. simple interest) so the demo can approve loans. This is **not** Alice’s official office formula. Do not draw it as a confirmed business rule. Write “demo placeholder pending client confirmation” on any figure that mentions rates.

2. People on the paper

Person	Role on the paper	You work with them how
Min Chen	Build and Railway	Technical questions only. He reviews Friday.
Myo Thant	Coordinator	Send all files to Myo. He keeps time and packs the report.
Ian Louw (you)	Visio + report prose	Figures 1–6 this week; 7–10 next week.
Asinate Lolohea	Client + 3 customer pages	She talks to Alice. Do not wait for her to start your diagrams.
Khoi Anh Do Tran	Screenshot evidence	He photographs the live screen that matches each of your figures.

3. How many figures

Ten figures in total. Six this week. Four next week. No class diagram. No extra sequence diagram per button.

#	Figure	Week	Type in Visio
1	System context	2	Context / DFD level 0
2	Use cases	2	UML use case
3	Happy-path activity	2	UML activity
4	Default-path activity	2	UML activity
5	Application and loan states	2	UML state machine
6	Collateral asset states	2	UML state machine
7	Logical ERD	3	Entity-relationship
8	Deployment / architecture	3	Block / deployment
9	Sequence — sign in	3	UML sequence
10	Sequence — record a repayment	3	UML sequence

Permission matrix and page list are **tables**, not extra Visio files. Asinate’s three customer-page mockups (Help, customer home, application progress) are UI, not your set.

4. Exact statuses — copy these labels

Use the identifiers below. Do not add “Overdue” as a loan status (overdue is only a calculated warning). Do not add “Pending approval” — the application status is SUBMITTED.

Thing	Allowed values (in order)
Application	DRAFT → SUBMITTED → APPROVED or DECLINED
Application steps (inside DRAFT)	BORROWER → LOAN_DETAILS → SECURITY → TERMS → REVIEW
Loan	APPROVED_UNFUNDED → ACTIVE → SETTLED, or ACTIVE → DEFAULTED
Collateral asset	PROPOSED → VALUED → STORED → RETURNED or SOLD
Valuation	REQUESTED → IN_PROGRESS → COMPLETED
Custody events	RECEIVED, INSPECTED, STORED, RELOCATED, RETURNED, SOLD
Correction	REQUESTED → APPROVED or REJECTED → POSTED (if approved)

5. Actors

Actor	What they do in this system
Customer	Register, verify email, view own applications and loans, receipts, help.
Loan officer	Create borrower, draft application, attach collateral, submit for review.
Valuation officer	Record valuation; later return or sell stored collateral.
Manager	Approve or decline; declare default; decide correction requests.
Cashier	Disburse, take repayment, record sale proceeds, request corrections.
Owner / admin staff	Invite staff, view audit. Owner is not a super-cashier.
Office (Alice)	External. Supplies official rates later. Not a system user account.
Email adapter	External. Verification and invites. School demo does not need real SMTP.
PostgreSQL	External data store. Not an actor on the use-case diagram; it belongs on context and architecture.

Accountant exists as a role name in the design but has no separate money-posting screens. You may show it on the use-case diagram with a note “view only / not implemented as a cash role”, or omit it.

6. Figure-by-figure instructions

Figure 1 — System context

One box in the centre: **Toumu’s Lending Workspace**.

- Left/top people: Customer, Loan officer, Valuation officer, Cashier, Manager, Owner/admin.
- Right systems: PostgreSQL, Email adapter, Railway host (optional, or fold into the centre box).
- Office / Alice as an external information source (official rates), not a daily login.

Arrows are information, not every API call. Example labels: “application and payments”, “receipts and balances”, “invite / verify email”, “persist loans”.

Figure 2 — Use case diagram

Group use cases. Stick-figure actors outside the system boundary.

- *Identity*: Register, verify email, sign in, reset password, accept staff invitation.
- *Origination*: Maintain borrower, create application, add collateral, record valuation, submit, approve/decline.
- *Servicing*: Intake collateral, disburse, record repayment, issue receipt, correct a transaction.
- *Close*: Return collateral; declare default; record sale; record sale proceeds.
- *Customer portal*: View my loans, schedule, security photos, notifications.

Include relationships only where they are real: «include» submit → completeness checks; approve «extend» decline as two outcomes of “Decide application”.

Figure 3 — Activity diagram, happy path

Swimlanes: Loan officer | Valuation officer | Manager | Cashier | Customer (optional, for viewing).

Happy path only:

Create borrower → draft application (five steps) → valuation completed → submit → manager approves → collateral RECEIVED / INSPECTED / STORED → cashier disburses (loan becomes ACTIVE) → repayments until balance is 0.00 (SETTLED) → valuation officer returns collateral (RETURNED).

Show a decision after review: Approve / Decline. Decline is a stop, not the default path.

Figure 4 — Activity diagram, default path

Starts from an ACTIVE loan. Manager declares DEFAULTED (reason required). Valuation officer records a sale. Cashier records sale proceeds. Surplus or shortfall is reported. If the debt is fully recovered the loan becomes SETTLED; collateral is SOLD, not returned.

Do not invent an automatic “overdue = default” transition.

Figure 5 — Application and loan states

Two regions on one page is fine.

- Application: DRAFT → SUBMITTED → APPROVED | DECLINED. No arrow back from SUBMITTED except through a new application (do not draw “unsubmit”).
- Loan is created only on APPROVED: APPROVED_UNFUNDED → ACTIVE → SETTLED. From ACTIVE a manager may move to DEFAULTED. From DEFAULTED, full recovery may move to SETTLED.

Figure 6 — Collateral states

PROPOSED → VALUED → STORED → RETURNED (settled loan) or SOLD (after default). Custody events sit on the arrows (RECEIVED, INSPECTED, STORED, RETURNED, SOLD).

Figures 7–10 — Week 3 only (preview so you can plan)

ERD (logical). Main entities and the relationships. You do not need every column.

- User 1—* Session, Invitation, Notification, AuditEvent
- User *—* Borrower through BorrowerAccountLink
- Borrower 1—* Application, Loan
- Application 1—1 ApplicationTerms, 1—* ApplicationAsset, 0—1 Loan, 0—1 ApplicationDecision
- ApplicationAsset 1—* Valuation, AssetPhoto, CustodyEvent
- Loan 1—* ScheduleEntry, LedgerEntry, Receipt, PaymentAttempt
- LedgerEntry 0—* CorrectionRequest

Architecture. Customer browser and staff browser (and Electron) → Web (Railway) → API (Railway) → PostgreSQL. Secrets stay on the API. Desktop does not talk to the database.

Sequence — sign in. Browser → POST /auth/login → API checks password → session cookie (HttpOnly) → GET /auth/me.

Sequence — repayment. Cashier → idempotent POST repayment → API quotes allocation → writes ledger + updates schedule → receipt. Replay of the same key returns the original result.

7. Writing the captions

Under each figure write 4–8 sentences in English:

- What the figure shows.
- Which actors or statuses appear.
- One rule that must not be broken (example: “collateral cannot be returned while a balance remains”).
- If a rate or fee appears, mark it demo.

Myo will drop these captions into the weekly report. Put your name on the figure titles: “Figure 3. Happy-path origination and servicing (Ian Louw)”.

8. Sources you may copy from

- Live app: web-production-b5acc.up.railway.app
- GitHub (public): github.com/chenmin911130-cyber/toumua-lending
- Design screens: `design/v2/screens/` and `design/v2/index.html`
- States and API intent: `design/v2/IMPLEMENTATION_SPEC.md`
- Routes: `docs/routes.md`

You do not need to clone the repo if GitHub in the browser is enough. Do not request `.env` or Railway secrets.

Do not. Invent official interest, fees, LTV multiples, or holiday-calendar rules. Do not redraw all 42 UI screens (Khoi screenshots cover those). Do not add a family-sharing account, online card payment, or public self-serve application — those are out of scope. Do not mark the demo 21% rate as Alice’s policy.

9. Friday checklist

1. Figures 1–6 exported as PDF (and the Visio source if you have it).
2. Captions in a Word or Google Doc, English.
3. Send the pack to Myo. Khoi will attach screenshots with the matching route.
4. 20-minute talk-through on Friday. Expect two questions from Myo or Min.